

OIL INDIA LIMITED - eRTMAC OPERATIONS

OFFICIAL WELL COMPLETION REPORT (WCR) & DRILLING HISTORY
WELL IDENTIFIER: 31/2-10

1. Well Location & Technical Header

Operator: Oil India Limited / Joint North Sea	Well Name: 31/2-10
UTM Easting (X): 526916.9 m	UTM Northing (Y): 6740121.5 m
Latitude: 60.795595° N	Longitude: 3.494499° E
Total Depth (MD): 1829.1 m	Spud Depth: 456.2 m
Drilling Rig: eRTMAC Semi-Submersible Rig-07	Status: Plugged & Abandoned (P&A;) / Offset Reference

2. Stratigraphic Progression & Formation Tops

Formation / Stratum Name	Top Depth MD (m)	Lithology Characterization	Drillability / Risk Profile
Balder Fm.	456.2 m	Interbedded Sand / Shale	Loss / Depletion
Sele Fm.	608.7 m	Shale / Claystone with interbedded siltstone	Tight Hole / Swelling
Lista Fm.	761.3 m	Interbedded Sand / Shale	Loss / Depletion
Ekofisk Fm.	913.8 m	Chalk / Hard Calcareous Limestone	Fractured / Losses
Hardraade Fm.	1066.4 m	Interbedded Sand / Shale	Loss / Depletion
Draupne Fm.	1218.9 m	Interbedded Sand / Shale	Loss / Depletion
Sognefjord Fm.	1371.5 m	Interbedded Sand / Shale	Loss / Depletion
Heather Fm.	1524.0 m	Interbedded Sand / Shale	Loss / Depletion
Fensfjord Fm.	1676.6 m	Interbedded Sand / Shale	Loss / Depletion

3. Casing Design & Mud Program Summary

36 in Hole	30 in Conductor	350 m	Spud Mud / Seawater	1.05 SG	FIT 1.25 SG
26 in Hole	20 in Surface	1,050 m	KCl / Polymer	1.12 SG	LOT 1.45 SG
17-1/2 in Hole	13-3/8 in Interm.	2,200 m	Low Solids Non-Dispersed	1.24 SG	LOT 1.62 SG
12-1/4 in Hole	9-5/8 in Prod.	1529 m	Oil Based / Synthetic Mud	1.32 SG	LOT 1.78 SG
8-1/2 in Hole	7 in Liner	1829 m	Synthetic Oil Based (SOBM)	1.38 SG	LOT 1.88 SG

4. Historical Operational Incidents, NPT & Mitigations (Well: 31/2-10)

This section constitutes the institutional memory log extracted by the NWIS knowledge engine. All alerts generated during active drilling cite these exact records.

INCIDENT #1: Borehole Instability / Pack-off		
	Severity: HIGH	NPT Hours: 15.5 hrs
Depth Interval: 1280.0 m - 1340.0 m	Formation: HORDALAND GP	Status: Resolved on Site
Event Occurrence Narrative: Torque erratic with rapid pressure spikes (+600 psi SPP) indicating annular packing around drill collars at 1315m.		
Mitigation Action Applied: Discontinued drilling, pulled up 2 stands into shoe, pumped 40 bbl high-viscosity pill with 25 lb/bbl bentonite, circulated 2 bottoms-up until shale cavings cleared.		
Offset Drilling Recommendation & Institutional Lesson: Increase flow rate to 820 gpm to optimize annular cutting transport velocity in oversized top-hole sections.		